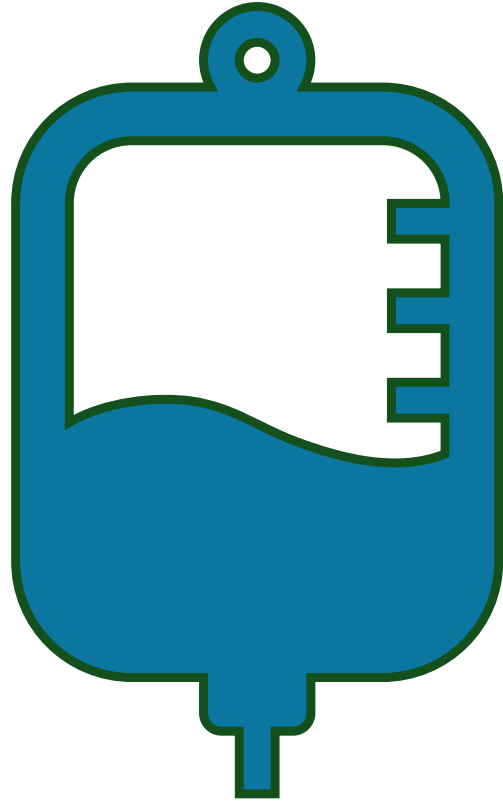




ELECTROLYTES

Chrissy McCain MSN, RN

OBJECTIVES

1. Identify assessment factors related to electrolyte and acid-base across age groups to include history, focused assessment, and diagnostics.
2. Identify therapeutic interventions related to electrolyte and acid-base across age groups to include pharmacologic, preventative, and collaborative.
3. Identify interrelated concepts related to electrolyte and acid-base concepts.
4. Describe common electrolyte and acid-base imbalances.
5. Interpret basic electrolyte and acid-base imbalances.
6. Describe signs and symptoms of common electrolyte and acid-base imbalances.
7. Apply the nursing process to common electrolyte and acid-base imbalances.

ELECTROLYTES

Sodium (Na)
Potassium (K)
Calcium (Ca)
Magnesium (Mag)
Phosphorous (Phos)
Chloride (Cl)

Your ultimate objective with electrolytes is to answer these questions:

- What is this electrolyte?
- What is its function; what does it do in the body?
- What body processes or tissues are affected by it?
- What is the normal value range? Critical values?
- What causes a low value?
- What signs & symptoms will you see when it is low? Critically low?
- What interventions will you implement to fix it?
- What causes a higher value?
- What signs & symptoms will you see when it is high? Critically high?
- What interventions will you implement to fix it?

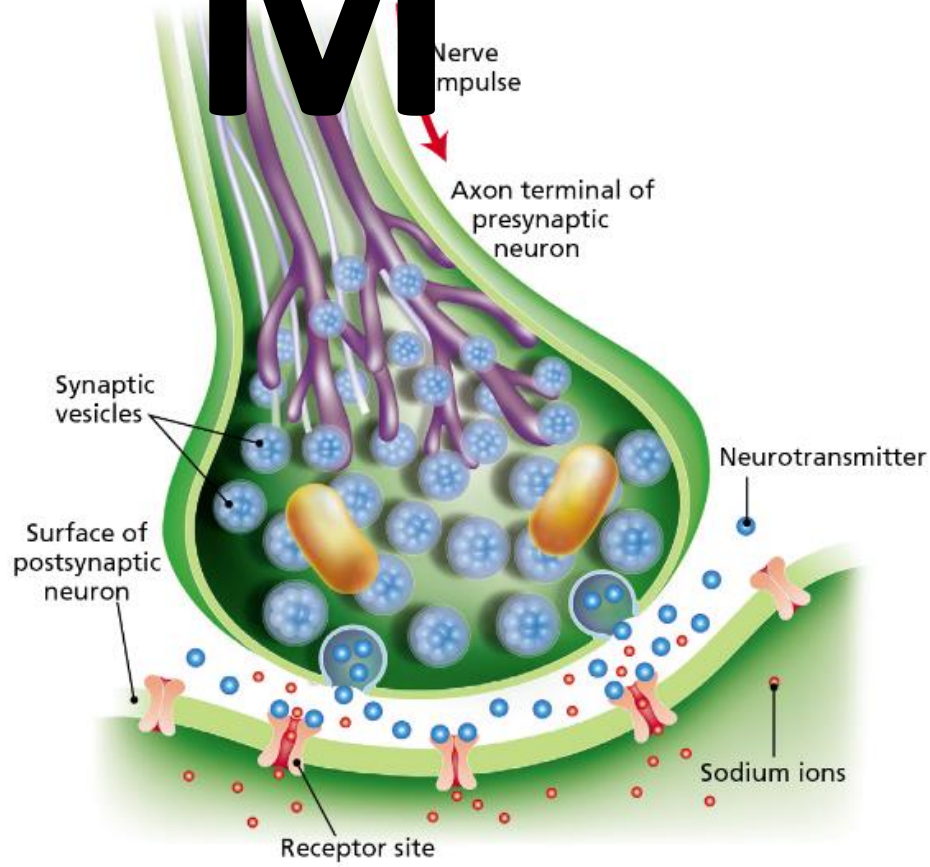
★ Interventions
also include what
we need to monitor
for due to S/S &
risks for...

ELECTROLYTES

- Electrolytes are part of a BMP or CMP (Basic -or- Complete Metabolic Panel) except for Magnesium.
- Nurses review lab results at beginning of shift or after a STAT-draw
- They identify abnormalities, assess patient's risk/stability, collaborate with the provider for medical interventions, and implement Nursing Interventions
- It is important for nurses to know what Nursing Interventions are appropriate and imperative including monitoring, increased vigilance, seizure precautions, collaboration with other disciplines, dietary needs, education needs, etc.

SODIU

M



RANGE: 135-145 mEq/L
GOAL: MAINTAIN BP + Blood Volume
MAINTAIN fluid balance
Regulated by Aldosterone

HYPONATREMIA → UNDER 135

CAUSES "NO Na⁺"

- > ↑ Na⁺ excretion = Renal problems, NG suction, vomiting, sweating, diuretics, DI (↓ Aldosterone secretion)
- > Overload of fluids = DILUTED
 - CHF, Hypotonic fluids, Liver Failure
- > Na⁺ intake is insufficient = Diet/NPO
- > ADH oversecretion = SIADH

INTERVENTIONS

1. Watch HR, RR, GI, Renal, Neuro
2. Administer IV Hypertonic Fluid → Hard on veins
3. Restrict Fluids / Diuretics Risk: fluid overload
4. Antidiuretic Hormone antagonist (for SIADH)
Desmopressin = Do not give with food!
5. Patients on Lithium = watch drug levels
6. DIET ↓ Na⁺ = ↑ Lithium

SIGNS + SYMPTOMS

W E A K A N D S H A K E Y

- > seizures/stupor
- > Abdominal cramping
- > Confusion/lethargic/trouble concentrating
- > ↓ DTR
- > Loss of urine and appetite
- > ↓ BP, ↓ Bowel sounds
- > Shallow respirations = LATE sign due to muscle weakness

- DIET -
ENCOURAGE

- > salty foods in moderation

HYPERNATREMIA → OVER 145

CAUSES

- > overproduction of Aldosterone
- > ↑ Na⁺ intake (oral/IV)
- > GI Tube feedings
- > Hypertonic solutions in excess
- > corticosteroids = Na⁺ excretion decreases
- > Loss of fluids = too much free salt
 - > Dehydrated, infection, diarrhea
- > Diabetes Insipidus

INTERVENTIONS

1. Restrict sodium intake
2. Patient safety (confused patient)
 - > CALL LIGHT
3. IV isotonic or hypotonic solution (cells are salty and shrunken)
 - > Give SLOWLY = RISK for cerebral edema
4. Educate on DIET

SIGNS + SYMPTOMS

B I G A N D B L O A T E D

- > Flushed skin
- > Fever
- > Agitated, confused
- > ↑ fluid Retention = EDEMA
- > ↓ urine output
- > Dry mouth + skin

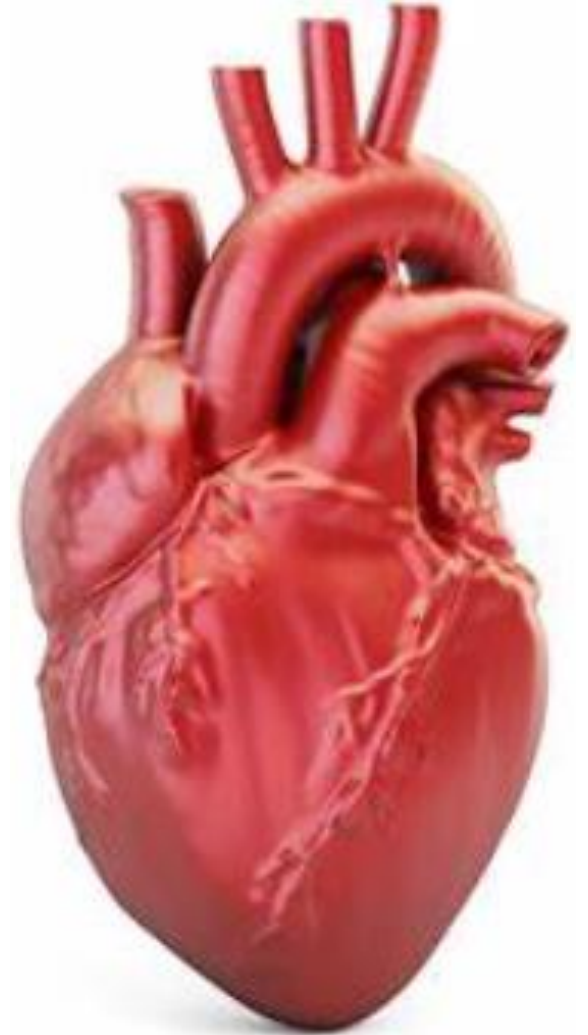
-DIET-

L I M I T

- | | |
|-----------------|---------------------|
| > canned foods | > cheese |
| > Frozen dinner | > salad dressing |
| > Fried food | > salty rice cakes |
| > Tomato sauce | > Pizza, hot dogs |
| | > chips, lunch meat |

POTASSIUM

RANGE: 3.5 - 5.0 mmol/L
GOAL: MANAGE heart + muscle function
MAINTAIN fluid balance + BP
REGULATED by kidneys



HYPOKALEMIA → UNDER 3.5

CAUSES

- > Dehydration
- > Diuretics (Furosemide)
- > Diarrhea/vomiting
- > Drains (NG Tubes)
- > severe acid imbalance
- > HYPERaldosteronism
- > corticosteroids → water retention
- insulin, albuterol → pushes K⁺ in the cell

INTERVENTIONS

1. Watch: EKG, respirations, GI, I & O's, BUN, creatinine
2. Watch Mg⁺ levels → K⁺ and Mg⁺ are bff's
3. Watch glucose, Ca⁺ and Na⁺ → Ca⁺, Na⁺, K⁺ are inversely related
4. give oral supplement WITH FOOD
5. < 2.5 = potassium infusion = ONLY IV SLOWLY
 > watch for infiltration
6. HOLD Potassium wasting diuretics! 

SIGNS + SYMPTOMS

LOW + SLOW

- > irregular, thready pulse
 - > ↓ BP, ↓ HR, orthostatic hypotension
 - > ↓ Bowel sounds
 - > ↓ DTR, flaccid paralysis, weak, leg cramps
 - > confusion
 - > shallow respiration, diminished breath sounds
 - > EKG: ↓ ST segment, inverted T wave, prominent U wave
- PEEING ALOT!

K⁺ sparing diuretics:

- > spironolone
- > aldactone
- > dyazide
- > triamterine
- > Maxide

HYPERKALEMIA → OVER 5.0

CAUSES

- > ACE Inhibitors (Retain K⁺)
- > Spironolactone
- > NSAIDs
- > Burns or trauma
(K⁺ explodes out of cells that are lysed)
- > Addison's = ↓ Aldosterone that regulates K⁺
- > IV Fluids
- > Renal impairment
(Kidney disease or dialysis)

INTERVENTIONS

- > IV sodium bicarbonate
- > IV calcium gluconate "gives the muscles down"
- > Albuterol
- > Furosemide / Hydrochlorothiazide
- > Dialysis
- > sodium polystyrene sulfonate

WATCH EKG ⚡

SIGNS + SYMPTOMS

TIGHT + CONTRACTED

- > Irregular heartbeat = ST elevation, peak T wave ⚡
- > ↓ BP, ↓ HR, severe VFib!
- > Respiratory Failure
- > Hyperactive bowel sounds = Diarrhea
- > Brain strain = confusion
- > BIG muscle weakness → cramping, ↓ DTR's
tingling, burning, numbness

LIMIT ⚡

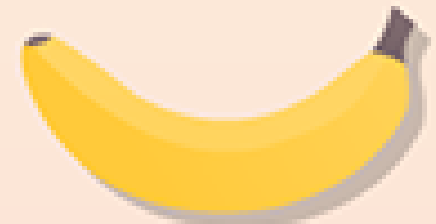
- DIET -

- No salt substitute
- No fruit
- No green leafy veggies

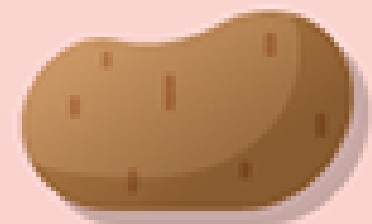
P - potatoes / pork
O - oranges
T - tomatoes
A - avocado
S - strawberries
S - spinach
fish
mushrooms + BANANAS ⚡
Melons

23 High Potassium Foods

					
Milk 407 mg 1 cup	Soy milk 300 mg 1 cup	Orange juice 220 mg ½ cup	Dried apricots 465 mg ¼ cup	Tomato sauce 455 mg ½ cup	Kidney beans 347 mg ½ cup
					
Yogurt 625 mg 1 cup	Soybeans 443 mg ½ cup	Banana 362 mg 1 small	Salmon 393 mg 3 ounces	Beef 257 mg 3 ounces	Raisins 297 mg ¼ cup
					
Chicken breast 282 mg 3 ounces	Avocado 364 mg ½ avocado	Acorn squash 448 mg ½ cup	Cantaloupe 243 mg 1 cup	Tomato 216 mg 1 small	Prunes 292 mg ¼ cup
					
Turkey breast 212 mg 3 ounces	Lentils 366 mg ½ cup	Potatoes 693 mg 1 small	Broccoli 288 mg 1 cup	Spinach 140 mg 1 cup	



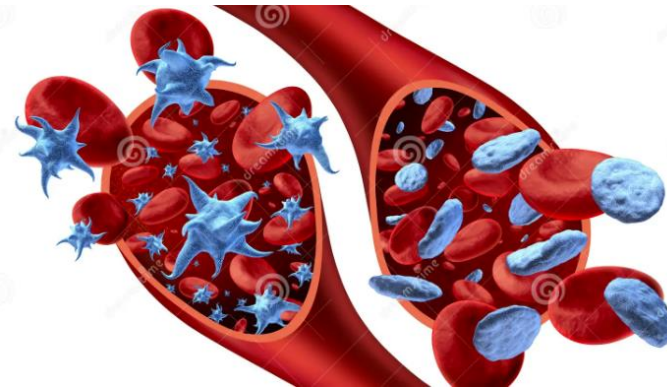
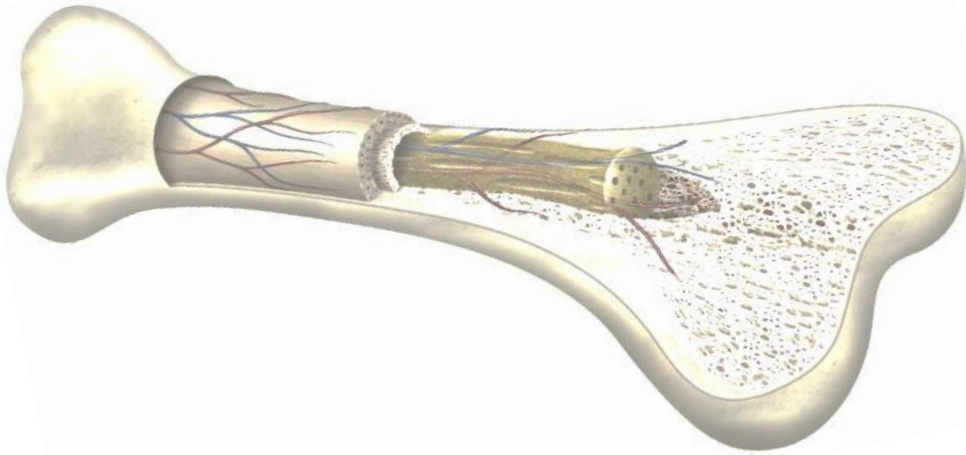
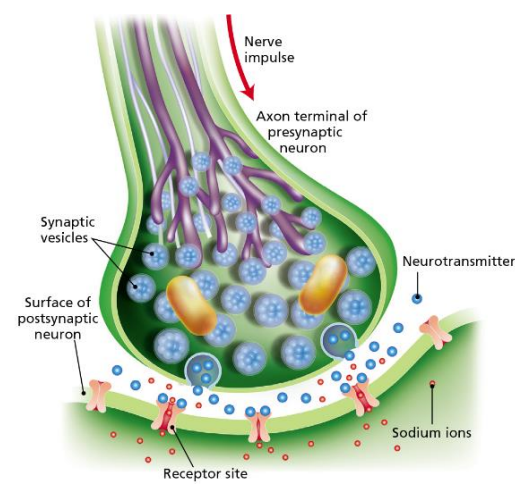
Banana
362 mg | 1 small



Potatoes
693 mg | 1 small

Kidney disease and some heart failure medications can affect your potassium levels.
Talk to your doctor or Livongo coach about any changes you may need to make to your diet when it comes to eating potassium-rich foods.

cALCIUM



RANGE: 9-11

GOAL: affects bones, heart beats, and clotting factors

JOB: stabilize neuron excitability

HYPOCALCEMIA - UNDER 9.0

CAUSES

- > Hypoparathyroidism
- > Thyroidectomy
- > Pancreatitis → releases calcium soaps + calcitonin
"Puts a TON of calcium in the bone"
- > Loop diuretics, laxatives, long term steroids, Phenytoin
- > Phosphate enemas = ↑ Phos = ↓ Ca⁺ & Mg⁺
- > GI wounds
- > Chronic disease → celiac, crohns, CKD
- > ↑ Phos, ↓ Vitamin D, ↓ magnesium

INTERVENTIONS

1. Give food high in calcium
2. calcium acetate (watch phos levels)
3. IV calcium
4. Oral calcium with Vitamin D
5. Magnesium Hydroxide

SIGNS + SYMPTOMS

WILD + CRAZY

- > Prolonged QT/ST (severe: VTach)
- > Heart Failure
- > Trousseau sign = arm spasm with BP cuff
- > Chvostek sign = smile when touching temporal lobe
- > slow clotting factors = bleeding
- > GI system going crazy = diarrhea
- > Laryngospasms, dyspnea
- > ALOC, seizures, confused

Watch EKG's

ENCOURAGE

Dairy
sardines
canned salmon
Leafy greens
Tofu
Edamame

HYPERCALCEMIA - OVER: 11.0

CAUSES

- > Hyperparathyroidism
- > Antacids with calcium (TUMS)
- > Malignant cancer cells
- > Low phosphate (Fraternal Twins)

INTERVENTIONS

1. Loop diuretics
2. Monitor EKG + urine output
3. IV normal saline for kidney stones
4. IV phosphate (Fraternal Twin)
5. calcium reabsorption inhibitors (calcitonin, Aspirin, NSAIDs)
6. Fall Risk
7. Last Resort = DIALYSIS
8. DIET

SIGNS + SYMPTOMS

S L O W + S W O L L E N

- > ↓ HR, ↓ RR, ↓ BP
- > short QT, wide Twave, ♥ muscle spasm
- > SOB, weak respirations
- > Hypoactive GI = constipation
- > Nausea, vomiting, ALOC
- > Renal calculi
- > ↓ DTR, ↓ muscle excitability
- if severe: muscle weakness
- > Bone pain: excess calcium was taken from bone

L I M I T Dairy
 sardines
 canned salmon
 Leafy greens
 Tofu
 Edamame

cALCIUM



Jarlsberg cheese
430 mg/44 g serve



Swiss cheese
410 mg/38 g serve



Edam cheese
340 mg/38 g serve



salmon
330 mg/120 g serve



gouda cheese
330 mg/42 g serve



cheddar cheese
330 mg/43 g serve



parmesan cheese
320 mg/25 g serve



mozzarella
290 mg/38 g serve



bok choy
16 g/2000 cal



arugula
13 g/2000 cal



spinach
12 g/2000 cal



beet greens
11 g/2000 cal



mackerel
240 mg/92 g serve



natto
220 mg/100 g serve



feta cheese
210 mg/37 g serve



Greek yogurt
208 mg/100 g serve



dill pickles
9 g/2000 cal



kale
9 g/2000 cal



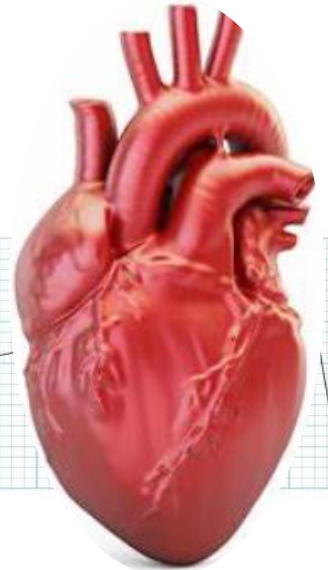
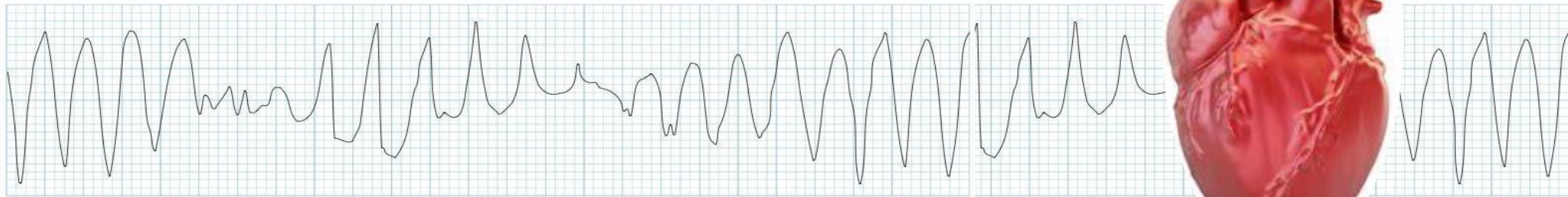
mustard greens
8 g/2000 cal



kimchi
7 g/2000 cal

mAGNESIuM

RANGE: 1.5 - 2.5 mEq/L
GOAL: muscle relaxation
MAINTAIN immune
system, bones, BG



HYPOMAGNESEMIA → UNDER 1.5

CAUSES

- > Excessive Alcohol = stops GI from absorbing Mg^{+}
- > Fluid loss → NG suction, N/V, Diuretics
- > Antibiotics (Aminoglycosides)
- > Pregnant mommas = risk for malnutrition

INTERVENTIONS

1. Assess swallowing (muscles)
2. IV magnesium sulfate
 - ↳ give slowly and monitor labs
3. Assess respiratory rate
4. Assess reflexes → CLONUS is BAD
5. DIET!

ENCOURAGE

Chocolate
Cauliflower
veggies
Avocados, Pears
Bananas, oranges
Milk, peanut butter
Pork, nuts

SIGNS + SYMPTOMS

MUSCLES GO WILD

- > ↑HR, ↑RR (shallow respirations)
- > Prolonged QT interval
- > Depressed ST segment, inverted Twave
- > Dyspnea
- > Diarrhea
- > ↑DTR's (CLONUS), numbness, tingling
- > Confusion, insomnia, seizures

BAD:

TORSADES DE POINTES
OR VFIB

HYPERMAGNESEMIA → OVER 2.5

CAUSES

- > Diabetic Ketoacidosis
- > Antacids with Mg^{+} (TUMS)
- > Renal Failure
- > Hyperkalemia - Addison's disease

INTERVENTIONS

1. IV Calcium gluconate
(will ↓ muscle tension + tightness)
2. Monitor labs and DTR's
3. Hemodialysis = takes out excess Mg^{+}
4. DIET!

SIGNS + SYMPTOMS

TOO RELAXED

- > ↓ BP, ↓ RR, ↓ HR
- > Widened QRS, Prolonged PR interval
- > Hypoactive bowel sounds
- > ↓ DTR's or absent
- > Drowsy, Lethargic

Σ BAD: COMA
Σ

L I M I T

Chocolate
Cauliflower
veggies
Avocados, Pears
Bananas, oranges
Milk, peanut butter
Pork, nuts

PHOSPHATE

RANGE: 2.5 - 4.5 mg/dL

GOAL: build bones, teeth and muscles

STORED: mainly in bones

REGULATED: by kidneys + parathyroid



HYPOPHOSPHATEMIA → UNDER 2.5

CAUSES

- > Antacids - malabsorption
- > Low vitamin D
- > Hyperparathyroidism = $\uparrow$ Ca⁺ & $\downarrow$ phos
- > Malignant cancer cells
 - ↳ kidneys waste phos = $\downarrow$ phos and bone softening
- > severe malnutrition
- > Hyperglycemia
- > Excessive alcohol, burns
- > Hypercalcemia

INTERVENTIONS

1. Give oral phosphate/vit D
2. IV phosphate → ensure kidney function
3. Fall Risk! watch calcium level
4. DIET! EKG

SIGNS + SYMPTOMS

BRITTLE + WEAK

- > muscle weakness (hint: lungs)
- > $\downarrow$ DTR's
- > $\downarrow$ cardiac output
- > Osteomalacia (Risk: fractures)
- > Immunosuppression = $\downarrow$ platelets, $\uparrow$ bleeding
- > Irritable, seizure RISK, confusion

ENCOURAGE

Fish, chicken
Nuts
Pork
Whole grains
Beans

HYPERPHOSPHATEMIA → OVER 4.5

CAUSES

- > Fleet OR sodium-phosphate enema
Why? ↑ phosphate levels d/t kidneys can't filter
- > Overuse of Vitamin D
- > Hypoparathyroidism = ↓ Ca^{2+} & ↑ phos
- > Insufficiency of kidneys (phos excreted)
- > Chemotherapy = kills good cells → electrolytes spill into blood

INTERVENTIONS

1. ⚡ GIVE PHOSLO = calcium acetate (WITH FOOD!)
2. Avoid phosphate enemas
3. DIET!
4. Last resort = DIALYSIS

SIGNS + SYMPTOMS

SAME AS HYPOCALCEMIA

- > Trousseau sign = arm spasm with BP cuff
- > Chvostek sign = smile when touching temporal lobe
- > Muscle spasms (TETANY) in calves + feet
- > Hyperactive DTR's, Bone pain
- > Laryngospasms
- > Confused / mental status changes

L I M I T

Fish, chicken
Nuts
Pork
Whole grains
Beans

ELECTROLYTE RELATIONSHIPS

Calcium & Phosphate

Low Ca = High Phos S&S

High CA = Low Phos S&S

*Remember Chvostek's &
Trousseau's Signs

Magnesium & Calcium

When *LOW* = high spasticity